

$$\begin{array}{l}
 \text{Row 2} = \text{Row 2} \rightarrow \begin{bmatrix} 3 & -7 & 0 & 0 & -19 \\ 0 & 1 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} \\
 \times (-1/2) \rightarrow \begin{bmatrix} 3 & 0 & 0 & 0 & -11 \\ 0 & 1 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} \\
 \text{Row 1} = \text{Row 1} \rightarrow \begin{bmatrix} 3 & 0 & 0 & 0 & -11 \\ 0 & 1 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} \\
 + 7 \times \text{Row 2} \rightarrow \begin{bmatrix} 3 & 0 & 0 & 0 & -11 \\ 0 & 1 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} \\
 \text{Row 1} = \text{Row 1} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 5/2 \\ 0 & 1 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} \\
 \times 1/3 \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 5/2 \\ 0 & 1 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} \\
 \text{Row 1} = \text{Row 1} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 & -6 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} \\
 \times 1/3 \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 & -6 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} \\
 \therefore x = \begin{bmatrix} 3 \\ 4 \\ -6 \\ -1 \end{bmatrix} \quad \square
 \end{array}$$

An LU Factorization Algorithm

Suppose A can be reduced to an echelon form U using only row replacements that add a multiple of one row to another row below it. In this case, there exist unit lower triangular elementary matrices $E_1, \dots, E_p$ such that

$$E_p \dots E_1 A = U \quad (3)$$

$$\text{Then } A = (E_p \dots E_1)^{-1} U = LU \quad \text{where } L = (E_p \dots E_1)^{-1} \quad (4)$$

We shall show in Exercise 19 that products and inverses of unit lower triangular matrices are also unit lower triangular.

$\therefore L$ is unit Lower Triangular.

Note that the row operations in equation (3), which reduce A to U , also reduce the L in equation (4) to I , because $E_p \dots E_1 L = (E_p \dots E_1)$. $(E_p \dots E_1)^{-1} = I$. This observation is the key to constructing L .

ALGORITHM FOR AN LU FACTORIZATION

1. Reduce A to an echelon form U by a sequence of row replacement operations, if possible.
2. Place entries in L such that the same sequence of row operations reduces L to I .